

PROJECT DOCUMENTATION

PROJECT PLANNING PHASES

EduGenie – Google Gemini Powered Learning Assistant

1. Project Definition

Define EduGenie as a Google Gemini Powered Learning Assistant for academic question answering, concept revision, self-assessment, quick revision, and topic-wise learning plans.

2. Objective Planning

Plan the development of a simple AI educational assistant, Gemini integration, concise answers, concept simplification, quiz generation, summarization, and structured learning guidance.

3. Technology Planning

Plan the use of Python, FastAPI, Uvicorn, Google Gemini / Google GenAI SDK, Jinja2, HTML5, CSS3, and python-dotenv.

4. Module Planning

Plan Question Answering, Concept Explanation, Quiz Generation, Summarization, and Personalized Learning Path modules.

5. Implementation Planning

Organize the project files as main.py, .env, .gitignore, templates/index.html, and static/style.css.

6. Testing Planning

Prepare tests for application loading, question answering, empty input validation, CSS loading, Gemini connection, and server startup.

7. Demonstration Planning

Prepare the project interface, module screens, test example, and output documentation for final demonstration.

8. Future Planning

Plan possible extensions including voice, multilingual learning, mobile applications, LMS integration, accounts, progress tracking, analytics, adaptive recommendations, and teacher/admin dashboards.

Prepared for project documentation and academic demonstration.